

André M. Silva

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Short Bio

I am André Silva, a postdoctoral researcher at the Instituto de Astrofísica e Ciências do Espaço. My research spans instrumentation and exoplanetary science. I am currently working on the PoET solar telescope, leading the development of its observation software. In parallel, I focus on the development of novel algorithms for precise radial velocity extraction and analyze systematic effects that impact such measurements. Over the past three years, I have lectured an advanced course on Python, a Master's-level course on Bayesian statistics, and a PhD-level module on high-resolution spectroscopy.

Education

PhD	Faculdade de Ciências , Doctor's Degree in Astronomy Thesis - A new paradigm for the estimation of precise stellar radial velocities	Universidade do Porto 2019 – 2024
MSc	Faculdade de Ciências , Mestrado Integrado em Engenharia Física Thesis - An expansion to the CHEOPS mission official pipeline.	Universidade do Porto 2014 – 2019

Awards

IAU PhD Prize 2024 🔗 , Division B	2025
Awarded the PhD Prize 2024 from the International Astronomical Union (IAU) Division B (Facilities, Technologies, and Data Science) for the work developed during my PhD thesis.	
EAS MERAC 2026 🔗 , Division B	2025
Awarded the European Astronomical Society (EAS) MERAC Prize in Observational Astrophysics for the work developed during my PhD thesis.	

Experience

Instituto de Astrofísica e Ciências do Espaço , Post-doctoral researcher Development of the observational software for the PoET solar telescope.	CAUP Nov 2024 – present
Faculdade Ciências da Universidade do Porto , Invited Assistant Professor Invited assistant Professor to teach a PhD-level unit on high-resolution spectroscopy	Physics department Nov 2024 – present
Faculdade de Ciências da Universidade de Lisboa , Post-doctoral researcher Development of software to measure day-time seeing, under the FIERCE ERC project	FCiências.ID Aug 2024 – Nov 2024
Faculdade Ciências da Universidade do Porto , Invited assistant Invited assistant for the practical classes of MSc-level unit on Bayesian data analysis.	Physics department Feb 2024 – Sept 2024
Instituto de Astrofísica e Ciências do Espaço , PhD fellow FCT funding for the PhD thesis "A new paradigm for the estimation of precise stellar radial velocities"	Physics department 2019 – July 2024

Talks

Physis Week 2025 , Python for data analysis	Sept 2025
ENAA XXXV 🔗 , A Quest to Detect a Star Velocity: Radial Velocity Extraction and Associated Challenges	Sept 2025
EPRV6 🔗 , A systematic bias in template-based RV extraction algorithms	July 2025

Planets throughout the Habitable Zone ↗ , A systematic bias in template-based RV extraction algorithms	June 2025
PoET workshop 2 ↗ , 2 talks: PoET observation software; Observing strategies and their role in the improvement of RV analysis	Nov 2024
TOI III ↗ , Approaches for RV extraction – s-BART and the first steps towards a fully Bayesian model	July 2023
EPRV 5 ↗ , Towards a fully Bayesian RV extraction model	Mar 2023
Exoplanets 4 EPRV splinter , sBART application to the ESPRESSO WG1 targets	Feb 2023
ESPRESSO science team meeting , s BART a semi Bayesian implementation of template matching for precise Radial Velocities	May 2022

Seminars

Astronomical Institute of the Czech Academy of Sciences , Avenues and challenges for radial velocity extraction – s-BART & PoET	Aug 2025
Astronomy department, Uni. of Geneva , Avenues for radial velocity extraction – s-BART & PoET	Jan 2025
ESA research seminar, European Space Agency (Madrid) , Towards an improvement in the characterisation of stellar radial velocities	May 2024
Café com Física ↗ Departamento de Física da Universidade de Coimbra , The (radial) velocity of stars - detection and characterisation of exoplanets	Mar 2024

Supervision

Clustering of spectral lines based on their parameters , Co-supervisor	MsC thesis Feb 2026 – present
Improved modelling of CCF profiles for activity-insensitive radial-velocity extraction , Supervisor	Undergraduate project Feb 2026 – present
Measuring Zeeman Broadening in Stellar Spectra , Co-supervisor	Undergraduate project July 2026 – present
Transforming H-alpha images of the Sun in astronomical seeing for the PoET solar telescope , Supervisor	Undergraduate project Feb 2025 – July 2025
Looking at the Sun, finding other Earths - Identification of Solar Regions , Co-supervisor	MsC thesis Oct 22 - Nov 23
Development of a GUI interface for the exploitation of data from the s-BART pipeline , Co-supervisor	Undergraduate project Feb 2024 – July 2024
Solar-to-sky coordinate conversion for PoET operations , Co-supervisor	Undergraduate project Feb 2024 – July 2024
Exoplanet detection through a new model for the correction of instrumental effects , Co-supervisor	Undergraduate project Feb 2024 – July 2024
Development of a tool for the normalization of stellar spectra - application to ESPRESSO data , Co-supervisor of BII project with ref CIAAUP-03/2023-BII ↗	Funded project Feb 2024 – July 2024
A new activity proxy for finding other Earths , Co-supervisor	Bachelors project Nov 2022 – Feb 2023
An analysis of the performance of CHEOPS mission pipelines - the DRP and archi , Co-supervisor	Bachelors project Feb 2020 – June 2020

First Author Publications

- ASTRA: A Python Package for Cross-Instrument Stellar and Telluric Template Construction** 2026
André M. Silva et al; (JOSS, [10.21105/joss.09413](https://doi.org/10.21105/joss.09413) )
- A systematic bias in template-based radial velocity extraction algorithms** 2025
André M. Silva et al; (A&A, [10.1051/0004-6361/202554955](https://doi.org/10.1051/0004-6361/202554955) )
- A novel framework for semi-Bayesian radial velocities through template matching** 2022
André M. Silva et al; (A&A, [10.1051/0004-6361/202142262](https://doi.org/10.1051/0004-6361/202142262) )
- archi: pipeline for light curve extraction of CHEOPS background stars** 2020
André M. Silva et al; (Monthly Notices of the Royal Astronomical Society, [10.1093/mnras/staa1443](https://doi.org/10.1093/mnras/staa1443) )

Co-authored Publications

- Gas-depleted planet formation occurred in the four-planet system around the red dwarf LHS 1903** 2026
Wilson et al; (Science, [10.1126/science.adl2348](https://doi.org/10.1126/science.adl2348) )
- A comprehensive study on radial velocity signals using ESPRESSO: Pushing precision to the 10 cm/s level** 2025
Figueira et al; (A&A, [10.1051/0004-6361/202553869](https://doi.org/10.1051/0004-6361/202553869) )
- TOI-2322: Two transiting rocky planets close to the stellar rotation period and its first harmonic** 2025
Hobson et al; (A&A, [10.1051/0004-6361/202555614](https://doi.org/10.1051/0004-6361/202555614) )
- Synthesizing Sun-as-a-star flare spectra from high-resolution solar observations** 2025
De Wilde et al; (A&A, [10.1051/0004-6361/202554870](https://doi.org/10.1051/0004-6361/202554870) )
- Two neighbours of the ultra-short-period Earth-sized planet K2-157 b in the warm Neptunian savanna** 2025
Castro-González et al; (A&A, [10.1051/0004-6361/202554736](https://doi.org/10.1051/0004-6361/202554736) )
- PoET: the Paranal solar ESPRESSO Telescope** 2025
Santos et al; (Published in The Messenger vol. 194, [10.18727/0722-6691/5381](https://doi.org/10.18727/0722-6691/5381) )
- TOI-512: Super-Earth transiting a K-type star discovered by TESS and ESPRESSO** 2025
Rodrigues et al; (A&A, [10.1051/0004-6361/202452887](https://doi.org/10.1051/0004-6361/202452887) )
- TESS and HARPS-N unveil two planets transiting TOI-1453: A super-Earth and one of the lowest mass sub-Neptunes** 2025
Stalport et al; (A&A, [10.1051/0004-6361/202452969](https://doi.org/10.1051/0004-6361/202452969) )
- A Planet Candidate Orbiting near the Hot Jupiter TOI-2818 b Inferred through Transit Timing** 2025
McKee et al; (ApJ, [10.3847/1538-4357/adac63](https://doi.org/10.3847/1538-4357/adac63) )
- Revisiting the multi-planetary system of the nearby star HD 20794: Confirmation of a low-mass planet in the habitable zone of a nearby G-dwarf** 2025
Nari et al; (A&A, [10.1051/0004-6361/202451769](https://doi.org/10.1051/0004-6361/202451769) )
- A sub-Earth-mass planet orbiting Barnard's star: No evidence of transits in TESS photometry** 2025
Stefanov et al; (A&A, [10.1051/0004-6361/202452450](https://doi.org/10.1051/0004-6361/202452450) )
- KOBE-1: The first planetary system from the KOBE survey: Two planets likely residing in the sub-Neptune mass regime around a late K-dwarf** 2025
Balsalobre-Ruza et al; (A&A, [10.1051/0004-6361/202452631](https://doi.org/10.1051/0004-6361/202452631) )

Detecting and characterising exoplanets with HARPS-N	2024
Rice et al; (Proc. IAU, 10.1017/S1743921324001741)	
Characterization of K2-167 b and CALM, a new stellar activity mitigation method	2024
De Beurs et al; (Monthly Notices of the Royal Astronomical Society, 10.1093/mnras/stae207)	
Expanding the frontiers of cool-dwarf asteroseismology with ESPRESSO: Detection of solar-like oscillations in the K5 dwarf ϵ Indi	2024
Campante et al; (A&A, 10.1051/0004-6361/202449197)	
TESS and ESPRESSO discover a super-Earth and a mini-Neptune orbiting the K-dwarf TOI-238	2024
Suárez Mascareño et al; (A&A, 10.1051/0004-6361/202348958)	
Confronting compositional confusion through the characterisation of the sub-Neptune orbiting HD 77946	2024
Palethorpe et al; (Monthly Notices of the Royal Astronomical Society, 10.1093/mnras/stae707)	
The compact multi-planet system GJ 9827 revisited with ESPRESSO	2024
Passegger et al; (A&A, 10.1051/0004-6361/202348592)	
Implementation of a seeing measurement device for the PoET solar telescope	2024
Wehbé et al; (Proc. SPIE , 10.1117/12.3017481)	
PoET, the Paranal solar ESPRESSO Telescope: a spatially resolved Sun in a high resolution spectrograph	2024
Leite et al; (Proc. SPIE , 10.1117/12.3016776)	
ESPRESSO reveals blueshifted neutral iron emission lines on the dayside of WASP-76 b	2024
Costa Silva et al; (A&A, 10.1051/0004-6361/202449935)	
A sub-Earth-mass planet orbiting Barnard's star	2024
González Hernández et al; (A&A, 10.1051/0004-6361/202451311)	
Two temperate Earth-mass planets orbiting the nearby star GJ1002	2023
Mascareño et al; (A&A, 10.1051/0004-6361/202244991)	
An unusually low-density super-Earth transiting the bright early-type M-dwarf GJ 1018 (TOI-244)	2023
Castro-González et al; (A&A, 10.1051/0004-6361/202346550)	
KOBESim: A Bayesian observing strategy algorithm for planet detection in radial velocity blind-search surveys	2023
Balsalobre-Ruza et al; (A&A, 10.1051/0004-6361/202243938)	
Automatic model-based telluric correction for the ESPRESSO data reduction software: Model description and application to radial velocity computation	2022
Allart et al; (A&A, 10.1051/0004-6361/202243629)	
A candidate short-period sub-Earth orbiting Proxima Centauri	2022
Faria et al; (A&A, 10.1051/0004-6361/202142337)	
The KOBE experiment: K-dwarfs Orbited By habitable Exoplanets: Project goals, target selection, and stellar characterization	2022
Lillo-Box et al; (A&A, 10.1051/0004-6361/202243898)	
HD22496b: the first ESPRESSO standalone planet discovery	2021
Lillo-Box et al; (A&A, 10.1051/0004-6361/202141714)	

Posters

The Paranal solar ESPRESSO Telescope - towards a resolved view of the Sun	2024
André M. Silva et al, Leiden, Exoplanets 5	
A fully-Bayesian model for RV extraction	2024
André M. Silva et al, Leiden, Exoplanets 5	
Navigating Through a Sea of Activity – A Portuguese Solar Telescope	2024
André M. Silva et al, Lisbon, ENAA XXXV	
A Bayesian template matching approach applied to HARPS : towards the improvement of the RV precision	2021
André M. Silva et al, Online, European Astronomical Society Annual meeting 2021	
A semi-Bayesian implementation of template matching for precise Radial Velocities	2021
André M. Silva et al, Online, Statistical challenges in Modern astronomy VII	
A semi-Bayesian implementation of template matching for precise Radial Velocities	2021
André M. Silva et al, Online, Encontro Ciência 21	
ARCHI: pipeline for light curve extraction of CHEOPS background stars	2020
André M. Silva et al, Online, Europlanet Science Congress 2020	
A Bayesian approach to precise Radial Velocities	2020
André M. Silva et al, Online, 30th Encontro Nacional de Astronomia e Astrofísica	

Grants

Post-doctoral fellowship	Aug-Nov 2024
Software development for the PoET telescope, funded by the FIERCE ERC project, grant number 101052347, Faculdade de Ciências da Universidade de Lisboa	
PhD fellowship, Fundação para a Ciência e Tecnologia (FCT)	2021-2024
"A new paradigm for the estimation of precise stellar radial velocities - towards the development of an innovative data analysis software", Ref. 2020.05387.BD	
FLAD grant	2023
Fund travel to the EPRV V conference in Santa Bárbara, California, PAPERS 4 USA, Ref. 2023/052	
Research fellowship	Nov-Dec 2019
field of Planetary Systems at Instituto de Astrofísica e Ciências do Espaço, Ref. IA2019-17-BIM	
Scientific Initiation Stundenship	Apr-Sep 2019
field of Computacional Astrophysics at Instituto de Astrofísica e Ciências do Espaço, Ref. IA2019-04-BIC	

Skills

Programming: Proficient with Python and Git; Familiar with code optimization to large datasets through numpy operations and Cython development
Data manipulation: Proficient with numpy and pandas for the manipulation of large datasets, coupled with matplotlib for data visualization
Data Analysis: Analysis of complex datasets using classical statistics (scipy) and Bayesian models (emcee, tinyGP)
Languages: English (fluent), Portuguese (native)

Organization

Coordinator of internal working group focused on modelling time-series for exoplanet detection and characterisation	2025 - now
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Organizer of the scientific seminars of CAUP	September 2025 - now
Co-organizer of PoET's Working Group 1 - Radial velocities	2025 - now
Coordinator and lecturer of advanced course "Python for astronomers"	2021 - now
SoC of Dias da Física 2025	2025
Part of the Local Organization committee of EPRV6	2025

Refereed

Main referee of the MsC thesis of Telmo Monteiro, in FCUP	2025
Refereed 1 manuscript for A&A	

Outreach

Outreach talk in Comicon 2026; "PoET - um telescópio solar português"	Apr 2026
Ignite sessions; 'Torres Vedras 🔗 Ílhavo 🔗 Armamar'	2023-2024
Outreach talks to highschool students: Espaço vai à Escola - Descoberta de outra Terra - deteção de planetas fora do sistema solar	2022-2025
2022 (3 talks); 2023 (5 online; 5 in-person); 2024 (6 online; 4 in-person); 2025 (6 in-person; 17 online)	